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Structure and method for forming capacitors for a three-dimensional NAND

Abstract

Embodiments of a three-dimensional capacitor for a memory device and fabrication methods are disclosed. The method includes forming, on a first side of a first substrate, a peripheral circuitry having a plurality of peripheral devices, a first interconnect layer, a deep well and a first capacitor electrode. The method also includes forming, on a second substrate, a memory array having a plurality of memory cells and a second interconnect layer, and bonding the first interconnect layer of the peripheral circuitry with the second interconnect layer of the memory array. The method further includes forming, on a second side of the first substrate, one or more trenches inside the deep well, disposing a capacitor dielectric layer on sidewalls of the one or more trenches, and forming capacitor contacts on sidewalls of the capacitor dielectric layer inside the one or more trenches.

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Background/Summary

INCORPORATION BY REFERENCE (1) This application is a divisional of U.S. patent application Ser. No. 16/729,818, filed on Dec. 30, 2019 and titled “Structure and Method for Forming Capacitors for a Three-Dimensional NAND,” which claims priority to PCT/CN2019/095069 filed on Jul. 8, 2019, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) The present disclosure generally relates to the field of semiconductor technology, and more particularly, to a method for forming a three-dimensional (3D) memory.

BACKGROUND

(2) As memory devices are shrinking to smaller die size to reduce manufacturing cost and increase storage density, scaling of planar memory cells faces challenges due to process technology limitations and reliability issues. A three-dimensional (3D) memory architecture can address the density and performance limitation in planar memory cells.

(3) In a conventional 3D memory, operation of memory cells need high voltage and capacitors are usually implemented as voltage booster. Currently integrated circuits for a 3D memory mainly use capacitors such as metal-oxide-silicon (MOS) capacitors, metal-oxide-metal (MOM) capacitors or polysilicon-oxide-polysilicon (POP) capacitors. As the development of 3D memory (e.g., 3D NAND flash memory) towards high density and high capacity memory cells, the number of devices (e.g., transistors) and the number of metal wirings are continuously increasing. In the meantime, to reduce manufacturing cost, the area of a memory chip remains largely unchanged. Therefore, devices in a 3D memory chip have been scaled down to smaller and smaller dimensions. Because capacitance is proportional to the area of a capacitor, a two-dimensional (2D) capacitor needs large silicon area in order to provide sufficient capacitance for the integrated circuitry of a 3D memory. To further increase capacitance, the thickness of the dielectric layer (e.g., silicon oxide) between the two electrodes of a capacitor can be thinned down. However, a capacitor with a very thin dielectric layer can suffer various reliability issues. Therefore, there is a need for a capacitor which can provide sufficiently large capacitance for the 3D memory within a reduced silicon area on a wafer.

BRIEF SUMMARY

(4) Embodiments of a three-dimensional (3D) capacitor structure for a memory device and methods for forming the same are described in the present disclosure.

(5) One aspect of the present disclosure provides a method for forming a 3D capacitor for a memory device, which includes forming, on a first side of a first substrate, a peripheral circuitry having a plurality of peripheral devices, a first interconnect layer, a deep well and a first capacitor

electrode. The first capacitor electrode is electrically connected with the deep well. The method also includes forming, on a second substrate, a memory array having a plurality of memory cells and a second interconnect layer. The method further includes bonding the first interconnect layer of the peripheral circuitry with the second interconnect layer of the memory array, such that at least one peripheral device of the peripheral circuitry is electrically connected with at least one memory cell of the memory array. The method also includes forming, on a second side of the first substrate, one or more trenches inside the deep well, where the first and second sides are opposite sides of the first substrate. The method further includes disposing a capacitor dielectric layer on sidewalls of the one or more trenches, and forming capacitor contacts on sidewalls of the capacitor dielectric layer inside the one or more trenches.

(6) In some embodiments, forming the 3D capacitor also includes thinning the first substrate from the second side after bonding the first and second interconnect layers. In some embodiments, thinning the first substrate includes exposing the deep well on the second side of the first substrate.

(7) In some embodiments, forming the 3D capacitor further includes disposing a capping layer on the second side of the first substrate prior to forming one or more trenches.

(8) In some embodiments, forming the 3D capacitor also includes forming a deep trench isolation to define an active area for the three-dimensional capacitor. In some embodiments, forming the deep trench isolation includes forming a through-silicon-trench penetrating through the first substrate and exposing a portion of the first interconnect layer, and disposing an insulating material inside the through-silicon-trench. In some embodiments, forming the deep trench isolation includes forming a through-silicon-trench penetrating through the first substrate prior to forming the one or more trenches, and exposing a portion of the first interconnect layer. In some embodiments, the through-silicon-trench has a width smaller than twice of a thickness of the capacitor dielectric layer.

(9) In some embodiments, forming capacitor contacts includes disposing a conductive material on the sidewalls of the capacitor dielectric layer inside the one or more trenches, and removing the conductive material outside the one or more trenches. In some embodiments, removing the conductive material outside the one or more trenches includes chemical mechanical polishing.

(10) In some embodiments, forming the 3D capacitor also includes forming a second capacitor electrode on the capacitor contacts on the second side of the first substrate.

(11) In some embodiments, the bonding of the first interconnect layer of the peripheral circuitry with the second interconnect layer of the memory array includes dielectric-to-dielectric bonding and metal-to-metal bonding at a bonding interface.

(12) Another aspect of the present disclosure provides a 3D capacitor for a memory device that includes a deep well formed on a second side of a first substrate. A first side of the first substrate, opposite of the second side, includes a plurality of peripheral devices and a first interconnect layer. The 3D capacitor also includes a first capacitor electrode electrically connected with the deep well. The 3D capacitor further includes one or more trenches inside the deep well, and a capacitor dielectric layer on sidewalls of the one or more trenches. The 3D capacitor also includes capacitor contacts on sidewalls of the capacitor dielectric layer inside the one or more trenches, and a second capacitor electrode disposed on the capacitor contacts.

(13) In some embodiments, the first interconnect layer on the first side of the first substrate is bonded with a second interconnect layer of a memory array on a second substrate, such that at least one peripheral device on the first substrate is electrically connected with at least one memory cell of the memory array.

(14) In some embodiments, the 3D capacitor also includes a deep trench isolation. The deep trench isolation penetrates through the first substrate and defines an active area for the three-dimensional capacitor.

(15) In some embodiments, the deep trench isolation is filled with an insulating material such as silicon oxide, silicon nitride or silicon oxynitride.

- (16) In some embodiments, the capacitor dielectric layer includes silicon oxide, silicon nitride or silicon oxynitride.
- (17) In some embodiments, the capacitor dielectric layer is a high-k dielectric material, including hafnium oxide, zirconium oxide, aluminum oxide, tantalum oxide, magnesium oxide, lanthanum oxide, or a combination of two or more thereof.
- (18) In some embodiments, the one or more trenches penetrate through the deep well and extend into the first interconnect layer.
- (19) In some embodiments, the one or more trenches penetrate through a portion of the deep well on the first substrate.
- (20) In some embodiments, the capacitor contacts on the sidewalls of the capacitor dielectric layer inside the one or more trenches include tungsten, copper, aluminum, titanium, nickel, cobalt, titanium nitride, tantalum nitride, or a combination of two or more thereof.
- (21) Other aspects of the present disclosure can be understood by those skilled in the art in light of the description, the claims, and the drawings of the present disclosure.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings, which are incorporated herein and form a part of the specification, illustrate embodiments of the present disclosure and, together with the description, further serve to explain the principles of the present disclosure and to enable a person skilled in the pertinent art to make and use the present disclosure.
- (2) FIG. 1 illustrates a schematic top-down view of an exemplary three-dimensional (3D) memory die, according to some embodiments of the present disclosure.
- (3) FIG. 2 illustrates a schematic top-down view of a region of 3D memory die, according to some embodiments of the present disclosure.
- (4) FIG. 3 illustrates a perspective view of a portion of an exemplary 3D memory array structure, in accordance with some embodiments of the present disclosure.
- (5) FIG. 4 illustrates a cross-sectional view of a peripheral circuitry, according to some embodiments of the present disclosure.
- (6) FIG. 5 illustrates a cross-sectional view of a memory array, according to some embodiments of the present disclosure.
- (7) FIG. 6 illustrates a cross-sectional view of a 3D memory device after bonding the peripheral circuitry and the memory array, according to some embodiments of the present disclosure.
- (8) FIGS. 7, 8 and 9A illustrate cross-sectional views of a 3D memory device at various process stages, according to some embodiments of the present disclosure.
- (9) FIGS. 9B and 9C illustrate a cross-sectional and top-down views of a region of a 3D memory device, according to some embodiments of the present disclosure.
- (10) FIGS. 9D and 9E illustrate a cross-sectional and top-down views of a region of a 3D memory device, according to some embodiments of the present disclosure.
- (11) FIGS. 9F and 9G illustrate top-down views of a region of a 3D memory device, according to some embodiments of the present disclosure.
- (12) FIGS. 10 and 11A illustrate cross-sectional views of a 3D memory device at various process stages, according to some embodiments of the present disclosure.
- (13) FIGS. 11B and 11C illustrate a cross-sectional and top-down views of a region of a 3D memory device, according to some embodiments of the present disclosure.
- (14) FIG. 12A illustrates a cross-sectional view of a 3D memory device at a certain process stage, according to some embodiments of the present disclosure.
- (15) FIGS. 12B and 12C illustrate a cross-sectional and top-down views of a region of a 3D

memory device, according to some embodiments of the present disclosure.

(16) FIG. 13 illustrate a flow diagram of an exemplary method for forming a memory device with a 3D capacitor, according to some embodiments of the present disclosure.

(17) The features and advantages of the present invention will become more apparent from the detailed description set forth below when taken in conjunction with the drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements. The drawing in which an element first appears is indicated by the leftmost digit(s) in the corresponding reference number.

(18) Embodiments of the present disclosure will be described with reference to the accompanying drawings.

DETAILED DESCRIPTION

(19) Although specific configurations and arrangements are discussed, it should be understood that this is done for illustrative purposes only. A person skilled in the pertinent art will recognize that other configurations and arrangements can be used without departing from the spirit and scope of the present disclosure. It will be apparent to a person skilled in the pertinent art that the present disclosure can also be employed in a variety of other applications.

(20) It is noted that references in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” “some embodiments,” etc., indicate that the embodiment described can include a particular feature, structure, or characteristic, but every embodiment can not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases do not necessarily refer to the same embodiment. Further, when a particular feature, structure or characteristic is described in connection with an embodiment, it would be within the knowledge of a person skilled in the pertinent art to affect such feature, structure or characteristic in connection with other embodiments whether or not explicitly described.

(21) In general, terminology can be understood at least in part from usage in context. For example, the term “one or more” as used herein, depending at least in part upon context, can be used to describe any feature, structure, or characteristic in a singular sense or can be used to describe combinations of features, structures or characteristics in a plural sense. Similarly, terms, such as “a,” “an,” or “the,” again, can be understood to convey a singular usage or to convey a plural usage, depending at least in part upon context. In addition, the term “based on” can be understood as not necessarily intended to convey an exclusive set of factors and may, instead, allow for existence of additional factors not necessarily expressly described, again, depending at least in part on context.

(22) It should be readily understood that the meaning of “on,” “above,” and “over” in the present disclosure should be interpreted in the broadest manner such that “on” not only means “directly on” something, but also includes the meaning of “on” something with an intermediate feature or a layer therebetween. Moreover, “above” or “over” not only means “above” or “over” something, but can also include the meaning it is “above” or “over” something with no intermediate feature or layer therebetween (i.e., directly on something).

(23) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, can be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or process step in addition to the orientation depicted in the figures. The apparatus can be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein can likewise be interpreted accordingly.

(24) As used herein, the term “substrate” refers to a material onto which subsequent material layers are added. The substrate includes a “top” surface and a “bottom” surface. The top surface of the substrate is typically where a semiconductor device is formed, and therefore the semiconductor

device is formed at a top side of the substrate unless stated otherwise. The bottom surface is opposite to the top surface and therefore a bottom side of the substrate is opposite to the top side of the substrate. The substrate itself can be patterned. Materials added on top of the substrate can be patterned or can remain unpatterned. Furthermore, the substrate can include a wide array of semiconductor materials, such as silicon, germanium, gallium arsenide, indium phosphide, etc. Alternatively, the substrate can be made from an electrically non-conductive material, such as a glass, a plastic, or a sapphire wafer.

(25) As used herein, the term “layer” refers to a material portion including a region with a thickness. A layer has a top side and a bottom side where the bottom side of the layer is relatively close to the substrate and the top side is relatively away from the substrate. A layer can extend over the entirety of an underlying or overlying structure, or can have an extent less than the extent of an underlying or overlying structure. Further, a layer can be a region of a homogeneous or inhomogeneous continuous structure that has a thickness less than the thickness of the continuous structure. For example, a layer can be located between any set of horizontal planes between, or at, a top surface and a bottom surface of the continuous structure. A layer can extend horizontally, vertically, and/or along a tapered surface. A substrate can be a layer, can include one or more layers therein, and/or can have one or more layer thereupon, thereabove, and/or therebelow. A layer can include multiple layers. For example, an interconnect layer can include one or more conductive and contact layers (in which contacts, interconnect lines, and/or vertical interconnect accesses (VIAs) are formed) and one or more dielectric layers.

(26) In the present disclosure, for ease of description, “tier” is used to refer to elements of substantially the same height along the vertical direction. For example, a word line and the underlying gate dielectric layer can be referred to as “a tier,” a word line and the underlying insulating layer can together be referred to as “a tier,” word lines of substantially the same height can be referred to as “a tier of word lines” or similar, and so on.

(27) As used herein, the term “nominal/nominally” refers to a desired, or target, value of a characteristic or parameter for a component or a process step, set during the design phase of a product or a process, together with a range of values above and/or below the desired value. The range of values can be due to slight variations in manufacturing processes or tolerances. As used herein, the term “about” indicates the value of a given quantity that can vary based on a particular technology node associated with the subject semiconductor device. Based on the particular technology node, the term “about” can indicate a value of a given quantity that varies within, for example, 10-30% of the value (e.g., $\pm 10\%$, $\pm 20\%$, or $\pm 30\%$ of the value).

(28) In the present disclosure, the term “horizontal/horizontally/lateral/laterally” means nominally parallel to a lateral surface of a substrate, and the term “vertical” or “vertically” means nominally perpendicular to the lateral surface of a substrate.

(29) As used herein, the term “3D memory” refers to a three-dimensional (3D) semiconductor device with vertically oriented strings of memory cell transistors (referred to herein as “memory strings,” such as NAND strings) on a laterally-oriented substrate so that the memory strings extend in the vertical direction with respect to the substrate.

(30) Various embodiments in accordance with the present disclosure provide structures and fabricating methods for vertical capacitors with higher density and less consumption of silicon area on a wafer. By using vertical capacitors instead of conventional two-dimensional (2D) capacitors, the overall memory density and manufacturing cost of a 3D NAND flash memory can be improved.

(31) FIG. 1 illustrates a top-down view of an exemplary three-dimensional (3D) memory device **100**, according to some embodiments of the present disclosure. The 3D memory device **100** can be a memory chip (package), a memory die or any portion of a memory die, and can include one or more memory planes **101**, each of which can include a plurality of memory blocks **103**. Identical and concurrent operations can take place at each memory plane **101**. The memory block **103**, which can be megabytes (MB) in size, is the smallest size to carry out erase operations. Shown in FIG. 1,

the exemplary 3D memory device **100** includes four memory planes **101** and each memory plane **101** includes six memory blocks **103**. Each memory block **103** can include a plurality of memory cells, where each memory cell can be addressed through interconnections such as bit lines and word lines. The bit lines and word lines can be laid out perpendicularly (e.g., in rows and columns, respectively), forming an array of metal lines. The direction of bit lines and word lines are labeled as “BL” and “WL” in FIG. **1**. In this disclosure, memory block **103** is also referred to as a “memory array” or “array.” The memory array is the core area in a memory device, performing storage functions.

(32) The 3D memory device **100** also includes a periphery region **105**, an area surrounding memory planes **101**. The periphery region **105** contains many digital, analog, and/or mixed-signal circuits to support functions of the memory array, for example, page buffers, row and column decoders and sense amplifiers. Peripheral circuits use active and/or passive semiconductor devices, such as transistors, diodes, capacitors, resistors, etc., as would be apparent to a person of ordinary skill in the art.

(33) It is noted that, the arrangement of the memory planes **101** in the 3D memory device **100** and the arrangement of the memory blocks **103** in each memory plane **101** illustrated in FIG. **1** are only used as an example, which does not limit the scope of the present disclosure.

(34) Referring to FIG. **2**, an enlarged top-down view of a region **108** in FIG. **1** is illustrated, according to some embodiments of the present disclosure. The region **108** of the 3D memory device **100** can include a staircase region **210** and a channel structure region **211**. The channel structure region **211** can include an array of memory strings **212**, each including a plurality of stacked memory cells. The staircase region **210** can include a staircase structure and an array of contact structures **214** formed on the staircase structure. In some embodiments, a plurality of slit structures **216**, extending in WL direction across the channel structure region **211** and the staircase region **210**, can divide a memory block into multiple memory fingers **218**. At least some slit structures **216** can function as the common source contact for an array of memory strings **212** in channel structure regions **211**. A top select gate cut **220** can be disposed in the middle of each memory finger **218** to divide a top select gate (TSG) of the memory finger **218** into two portions, and thereby can divide a memory finger into two programmable (read/write) pages. While erase operation of a 3D NAND memory can be carried out at memory block level, read and write operations can be carried out at memory page level. A page can be kilobytes (KB) in size. In some embodiments, region **108** also includes dummy memory strings for process variation control during fabrication and/or for additional mechanical support.

(35) FIG. **3** illustrates a perspective view of a portion of an exemplary three-dimensional (3D) memory array structure **300**, according to some embodiments of the present disclosure. The memory array structure **300** includes a substrate **330**, an insulating film **331** over the substrate **330**, a tier of lower select gates (LSGs) **332** over the insulating film **331**, and a plurality of tiers of control gates **333**, also referred to as “word lines (WLs),” stacking on top of the LSGs **332** to form a film stack **335** of alternating conductive and dielectric layers. The dielectric layers adjacent to the tiers of control gates are not shown in FIG. **3** for clarity.

(36) The control gates of each tier are separated by slit structures **216-1** and **216-2** through the film stack **335**. The memory array structure **300** also includes a tier of top select gates (TSGs) **334** over the stack of control gates **333**. The stack of TSG **334**, control gates **333** and LSG **332** is also referred to as “gate electrodes.” The memory array structure **300** further includes memory strings **212** and doped source line regions **344** in portions of substrate **330** between adjacent LSGs **332**. Each memory strings **212** includes a channel hole **336** extending through the insulating film **331** and the film stack **335** of alternating conductive and dielectric layers. Memory strings **212** also includes a memory film **337** on a sidewall of the channel hole **336**, a channel layer **338** over the memory film **337**, and a core filling film **339** surrounded by the channel layer **338**. A memory cell **340** can be formed at the intersection of the control gate **333** and the memory string **212**. The

memory array structure **300** further includes a plurality of bit lines (BLs) **341** connected with the memory strings **212** over the TSGs **334**. The memory array structure **300** also includes a plurality of metal interconnect lines **343** connected with the gate electrodes through a plurality of contact structures **214**. The edge of the film stack **335** is configured in a shape of staircase to allow an electrical connection to each tier of the gate electrodes.

(37) In FIG. **3**, for illustrative purposes, three tiers of control gates **333-1**, **333-2**, and **333-3** are shown together with one tier of TSG **334** and one tier of LSG **332**. In this example, each memory string **212** can include three memory cells **340-1**, **340-2** and **340-3**, corresponding to the control gates **333-1**, **333-2** and **333-3**, respectively. In some embodiments, the number of control gates and the number of memory cells can be more than three to increase storage capacity. The memory array structure **300** can also include other structures, for example, TSG cut, common source contact and dummy channel structure. These structures are not shown in FIG. **3** for simplicity.

(38) To achieve higher storage density, the number of vertical WL stacks of a 3D memory or the number of memory cells per memory string has been greatly increased, for example, from 24 stacked WL layers (i.e. 24 L) to 128 layers or more. To further reduce the size of a 3D memory, the memory array can be stacked on top of the peripheral circuitry or vice versa. For example, the peripheral circuitry can be fabricated on a first substrate and the memory array can be fabricated on a second substrate. Then the memory array and the peripheral circuitry can be connected through various interconnects by bonding the first and second substrates together. As such, not only the 3D memory density can be increased, but also communication between the peripheral circuitry and memory array can achieve higher bandwidth and lower power consumption since the interconnect lengths can be shorter through substrate (wafer) bonding. FIGS. **4-8**, **9A-9G**, **10**, **11A-11C**, **12A-12C** and **13** illustrate the structures and methods for forming a 3D memory device where peripheral circuitry is connected with memory array through wafer bonding, according to some embodiments of the present disclosure.

(39) With the increase in the density and performance of the 3D memory device, improvement in the peripheral circuitry is also needed to provide functional support for the memory array, for example, reading, writing and erasing the data of the memory cells. Among the peripheral devices, capacitors are used to regular voltages in a 3D memory device, for example, boosting voltage for erasing memory data. Accordingly, FIGS. **4-8**, **9A-9G**, **10**, **11A-11C**, **12A-12C** and **13** illustrate a 3D capacitor of a memory device at various process stages, according to some embodiments of the present disclosure.

(40) FIG. **4** illustrates a cross-section of an exemplary peripheral circuitry **400** of a 3D memory device according to some embodiments of the present disclosure. The peripheral circuitry **400** can include a first substrate **430**, where the first substrate **430** can include silicon (e.g., single crystalline silicon), silicon germanium (SiGe), germanium (Ge), silicon on insulator (SOI), germanium on insulator (GOI), gallium arsenide (GaAs), gallium nitride, silicon carbide, glass, III-V compound, any other suitable materials or any combinations thereof. In some embodiments, the first substrate **430** can be double-side polished prior to peripheral device fabrication. In this example, the first substrate **430** includes surfaces on the top and bottom sides (also referred to as a first side **430-1** and a second side **430-2**, or a front side and a backside, respectively) both polished and treated to provide a smooth surface for high quality semiconductor devices. The first and second sides are opposite sides of the first substrate.

(41) The peripheral circuitry **400** can include one or more peripheral devices **450** on a first side **430-1** of the first substrate **430**. The peripheral device **450** can be formed “on” the first substrate **430**, in which the entirety or part of the peripheral device **450** is formed in the first substrate **430** (e.g., below the top surface of the first substrate **430**) and/or directly on the first substrate **430**. The peripheral device **450** can include any suitable semiconductor devices, for example, metal oxide semiconductor field effect transistors (MOSFETs), bipolar junction transistors (BJTs), diodes, resistors, capacitors, inductors, etc. Among the semiconductor devices, p-type and/or n-type

MOSFETs (i.e., CMOS) are widely implemented in logic circuit design, and are used as examples for the peripheral device **450** in the present disclosure. In this example, the peripheral circuitry **400** is also referred to CMOS wafer **400**.

(42) A peripheral device **450** can be either a p-channel MOSFET or an n-channel MOSFET and can include, but not limited to, an active device region surrounded by shallow trench isolation (STI) **452**, a well **454** formed in the active device region with n-type or p-type doping, a gate stack **456** that includes a gate dielectric, a gate conductor and/or a gate hard mask. The peripheral device **450** can also include a source/drain extension and/or halo region (not shown in FIG. **4**), a gate spacer **458** and a source/drain **460** locating on each side of the gate stack. The peripheral device **450** can further include a silicide contact area (not shown) in the top portion of the source/drain. Other known devices can be also formed on the first substrate **430**. The structure and fabrication method of the peripheral device **450**, are known to those skilled in the art, and are incorporated herein for entirety.

(43) The STI **452** can be formed through patterning the substrate using lithography and etching, filling an insulating material and polishing the insulating material to form a coplanar surface on the first substrate **430**. An insulating material for STI can include silicon oxide, silicon oxynitride, TEOS, low-temperature oxide (LTO), high temperature oxide (HTO), silicon nitride, etc. An insulating material for STI **452** can be disposed using techniques such as chemical vapor deposition (CVD), physical vapor deposition (PVD), plasma-enhanced CVD (PECVD), low pressure chemical vapor deposition (LPCVD), high density plasma (HDP) chemical vapor deposition, rapid thermal chemical vapor deposition (RTCVD), metal organic chemical vapor deposition (MOCVD), atomic layer deposition (ALD), sputtering, thermal oxidation or nitridation, or combinations thereof. The forming of STI **452** can also include a high temperature annealing step to densify the disposed insulating material for improved electrical isolation. Other STI structures can be employed, as would be apparent to a person of ordinary skill in the art.

(44) The well **454** of the peripheral device **450** can include a p-type doping for n-channel MOSFET and an n-type doping for p-channel MOSFET, and is called p-well and n-well, respectively. The dopant profile and concentration of the well **454** affects the device characteristics of the peripheral device **450**. For MOSFET devices with low threshold voltage ($V_{sub.t}$), the well **454** can be doped with lower concentration, and can form low-voltage p-well or low-voltage n-well. For MOSFET with high $V_{sub.t}$, the well **454** can be doped with higher concentration, and can form high-voltage p-well or high-voltage n-well. In some embodiments, to provide electrical isolation from a p-type substrate, a deep n-well can be formed underneath a high-voltage p-well for an n-channel MOSFET with high $V_{sub.t}$. In some embodiments, a depth of the well **454** can be deeper than a depth of the STI **452**.

(45) The forming of an n-well can include any suitable n-type dopant, such as phosphorus, arsenic, antimony, etc., and/or any combination thereof. The forming of a p-well can include any suitable p-type dopant, for example boron. The dopant incorporation can be achieved through ion implantation followed by activation anneal, or through in-situ doping during epitaxy for the active device region.

(46) The gate stack **456** of the peripheral device **450** can be formed by a “gate first” scheme, where the gate stack **456** is disposed and patterned prior to source/drain formation. The gate stack **456** of the peripheral device **450** can also be formed by a “replacement” scheme, where a sacrificial gate stack can be formed first and then replaced by a high-k dielectric layer and a gate conductor after source/drain formation.

(47) In some embodiments, the gate dielectric can be made of silicon oxide, silicon nitride, silicon oxynitride, and/or high-k dielectric films such as hafnium oxide, zirconium oxide, aluminum oxide, tantalum oxide, magnesium oxide, or lanthanum oxide films, and/or combinations thereof. The gate dielectric can be disposed by any suitable methods such as CVD, PVD, PECVD, LPCVD, RTCVD, sputtering, MOCVD, ALD, thermal oxidation or nitridation, or combinations thereof.

(48) In some embodiments, the gate conductor can be made from a metal or metal alloy, such as tungsten, cobalt, nickel, copper, or aluminum, and/or combinations thereof. In some embodiments, the gate conductor can also include a conductive material, such as titanium nitride (TiN), tantalum nitride (TaN), etc. The gate conductor can be formed by any suitable deposition methods, for example, sputtering, thermal evaporation, e-beam evaporation, ALD, PVD, and/or combinations thereof.

(49) In some embodiments, the gate conductor can also include a poly-crystalline semiconductor, such as poly-crystalline silicon, poly-crystalline germanium, poly-crystalline germanium-silicon and any other suitable material, and/or combinations thereof. In some embodiments, the poly-crystalline material can be incorporated with any suitable types of dopant, such as boron, phosphorous, or arsenic, etc. In some embodiments, the gate conductor can also be an amorphous semiconductor with aforementioned materials.

(50) In some embodiments, the gate conductor can be made from a metal silicide, including $\text{WSi}_{\text{sub}.x}$, $\text{CoSi}_{\text{sub}.x}$, $\text{NiSi}_{\text{sub}.x}$, or $\text{AlSi}_{\text{sub}.x}$, etc. The forming of the metal silicide material can include forming a metal layer and a poly-crystalline semiconductor using similar techniques described above. The forming of metal silicide can further include applying a thermal annealing process on the deposited metal layer and the poly-crystalline semiconductor layer, followed by removal of unreacted metal.

(51) The gate spacer **458** can be formed through disposing an insulating material and then performing anisotropic etching. The insulating material for the gate spacer **458** can be any insulator, including silicon oxide, silicon nitride, silicon oxyntiride, TEOS, LTO, HTO, etc. The gate spacer **458** can be disposed using techniques such as CVD, PVD, PECVD, LPCVD, RTCVD, MOCVD, ALD, sputtering, or combinations thereof. The anisotropic etching of the gate spacer **458** includes dry etching, for example reactive ion etching (ME).

(52) A gate length L of the gate stack **456** between the source/drain **460** is an important feature of the MOSFET. During operation of a MOSFET, a top portion of the well underneath the gate stack **456** can conduct current from source to drain, and is so called channel of the MOSFET. The gate length L (also referred to as channel length) determines the magnitude of drive current of a MOSFET and is therefore scaled down aggressively for logic circuits. The gate length L can be less than about 100 nm. In some embodiments, the gate length can be in a range between about 5 nm to about 30 nm. Patterning of the gate stack with such a small dimension is very challenging, and can use techniques including optical proximity correction, double exposure and/or double etching, self-aligned double patterning, etc.

(53) In some embodiments, the source/drain **460** of the peripheral device **450** is incorporated with high concentration dopants. For n-type MOSFETs, the dopant for source/drain **460** can include any suitable n-type dopant, such as phosphorus, arsenic, antimony, etc., and/or any combination thereof. For p-type MOSFETs, the dopant for source/drain **460** can include any suitable p-type dopant, for example boron. The dopant incorporation can be achieved through ion implantation followed by dopant activation anneal. The source/drain **460** can be made of the same material as the first substrate **430**, for example, silicon. In some embodiments, the source/drain **460** of the peripheral device **450** can be made of a different material from the first substrate **430** to achieve high performance. For example, on a silicon substrate, the source/drain **460** for a p-type MOSFETs can include SiGe and the source/drain **460** for an n-type MOSFETs can be incorporated with carbon. The forming of the source/drain **460** with a different material can include etching back the substrate material in the source/drain area and disposing new source/drain material using techniques such as epitaxy. Doping for source/drain **460** can also be achieved through in-situ doping during epitaxy.

(54) The peripheral device **450** can also have an optional source/drain extension and/or halo region (not shown in FIG. 4) along each side of the gate stack **456**. The source/drain extension and/or halo region locates inside the active device region below the gate stack, and is implemented mainly for

improved short channel control for the peripheral device **450** with a channel length less than about 0.5 μm . The forming of the source/drain extension and/or halo region can be similar to the forming of the source/drain **460**, but may use different implantation conditions (e.g., dose, angle, energy, species, etc.) to obtain optimized doping profile, depth or concentration.

(55) The peripheral device **450** can be formed on the first substrate **430** with a planar active device region (as shown in FIG. 4), where the direction of MOSFET's channel and current flow is parallel to a surface of the first substrate **430**. In some embodiments, the peripheral device **450** can also be formed on the first substrate **430** with a 3D active device region, for example a so-called "FINFET" in a shape like a "FIN" (not shown), where the gate stack of the MOSFET is wrapped around the FIN, and the MOSFET's channel lies along three sides of the FIN (top and two sidewalls under the gate). The structure and methods for FINFET device are known to those skilled in the art and are not discussed further in present disclosure.

(56) In some embodiments, the peripheral circuitry **400** can include a peripheral interconnect layer **462** (or a first interconnect layer) on the first side **430-1**, above the peripheral devices **450**, to provide electrical connections between different peripheral devices **450** and external devices (e.g., power supply, another chip, I/O device, etc.). The peripheral interconnect layer **462** can include one or more interconnect structures, for example, one or more vertical contact structures **464** and one or more lateral conductive lines **466**. The contact structure **464** and conductive line **466** can broadly include any suitable types of interconnects, such as middle-of-line (MOL) interconnects and back-end-of-line (BEOL) interconnects. The contact structure **464** and conductive line **466** in the peripheral circuitry **400** can include any suitable conductive materials such as tungsten (W), cobalt (Co), copper (Cu), titanium (Ti), tantalum (Ta), aluminum (Al), titanium nitride (TiN), tantalum nitride (TaN), nickel, silicides (WSi.sub.x, CoSi.sub.x, NiSi.sub.x, AlSi.sub.x, etc.), metal alloys, or any combination thereof. The conductive materials can be deposited by one or more thin film deposition processes such as chemical vapor deposition (CVD), plasma-enhanced CVD (PECVD), physical vapor deposition (PVD), atomic layer deposition (ALD), electroplating, electroless plating, sputtering, evaporation, or any combination thereof.

(57) The peripheral interconnect layer **462** can further include an insulating layer **468**. The insulating layer **468** in the peripheral interconnect layer **462** can include insulating materials, for example, silicon oxide, silicon nitride, silicon oxynitride, doped silicon oxide (such as F-, C-, N- or H-doped oxides), tetraethoxysilane (TEOS), polyimide, spin-on-glass (SOG), low-k dielectric material such as porous SiCOH, silsesquioxan (SSQ), or any combination thereof. The insulating materials can be deposited by one or more thin film deposition processes such as CVD, PVD, PECVD, ALD, high-density-plasma CVD (HDP-CVD), sputtering, spin-coating, or any combination thereof.

(58) In FIG. 4, two conductive levels **470-1** and **470-2** (also referred to as "metal levels") are illustrated as an example, where each metal level **470** (e.g., **470-1** or **470-2**) include the contact structures **464** and the conductive lines **466**. The conductive lines **466** of the same metal level are located at the same distance from the first substrate **430**. The number of metal levels **470** for the peripheral circuitry **400** is not limited and can be any number optimized for the performance of the 3D memory.

(59) The peripheral interconnect layer **462** can be formed by stacking metal levels **470** from bottom to the top of the peripheral circuitry **400**. In the example of the peripheral circuitry **400** in FIG. 4, the bottom metal level **470-1** can be formed first and then the upper metal level **470-2** can be formed on top of the bottom metal level **470-1**. Fabrication processes of each metal level **470** can include, but not limited to, disposing a portion of the insulating layer **468** with a thickness required for the metal level, patterning the portion of the insulating layer **468** using photo lithography and dry/wet etching to form contact holes for the contact structures **464** and the conductive lines **466**, disposing conductive materials to fill the contact holes for the contact structures **464** and the conductive lines **466**, and removing excessive conductive materials outside the contact holes by

using planarization process such as chemical mechanical polishing (CMP) or reactive ion etching (ME).

(60) In some embodiments, peripheral circuitry **400** also includes one or more substrate contacts **472**, where the substrate contacts **472** provide electrical connections to the first substrate **430**. The substrate contact **472** can include one or more conductive levels **470** with multiple tiers of vertical contact structures **464** and lateral conductive lines **466**. In FIG. 4, substrate contact **472** with one tier of contact structure and conductive line is shown as an example, where the vertical contact structure of the substrate contact **472** extends through the insulating layer **468** and electrically contacts the first substrate **430**.

(61) In some embodiments, the topmost conductive lines **466** (e.g., **466-2** in FIG. 4) can be exposed as the top surface of the peripheral circuitry **400**, where the topmost conductive lines **466-2** can be directly connected with the conductive lines on another chip or an external device.

(62) In some embodiments, the topmost conductive lines **466-2** can be embedded inside the insulating layer **468** (as shown in FIG. 4), where the insulating material on top of the conductive lines **466** provide scratch protection during shipping or handling. Electrical connections to the topmost conductive lines **466** can be established later by forming metal VIAs, or simply by etching back the insulating layer **468** using dry/wet etching.

(63) The peripheral device **450**, however, is not limited to MOSFET. The structures of the other devices, for example diodes, resistors, capacitors, inductors, BJTs, etc., can be formed simultaneously during MOSFETs fabrication through different mask design and layout. To form devices other than MOSFETs, process steps can be added or modified in a MOSFET's process flow, for example, processes to obtain different dopant profiles, film thicknesses or material stacks, etc. In some embodiments, the peripheral device **450** other than MOSFET can also be fabricated with additional design and/or lithography mask levels to achieve specific circuit requirements.

(64) In some embodiments, a plurality of the peripheral devices **450** can be used to form any digital, analog, and/or mixed-signal circuits for the operation of the peripheral circuitry **400**. The peripheral circuitry **400** can perform, for example, row/column decoding, timing and control, reading, writing and erasing data of the memory array, etc.

(65) In some embodiments, a 3D capacitor can be formed for the peripheral circuitry **400**. For example, a deep well **455** can be formed in the first substrate **430** while forming the wells **454** for MOSFETs. The deep well **455** can be p-type doped or n-type doped. The n-type dopant can be phosphorus, arsenic, antimony, etc. The p-type dopant can be, for example, boron. The dopant incorporation can be achieved through ion implantation from the first side **430-1** of the first substrate **430**, followed by activation anneal. In some embodiments, the deep well **455** can be formed on the first side **430-1** of the first substrate **430** through epitaxy and in-situ doping. The implantation for the deep well **455** can be performed right before or after the implantation for the well **454**. The dopant activation anneal for deep well **455** can be performed simultaneously as that for the well **454**.

(66) In some embodiments, the deep well **455** can have a depth in a range between 1 μm to 5 μm . In some embodiments, the deep well **455** is highly doped. For example, the deep well **455** can be doped to $1 \times 10^{18} \text{ cm}^{-3}$ or higher, i.e., having a dopant concentration higher than $1 \times 10^{18} \text{ cm}^{-3}$.

(67) In some embodiments, a deep well contact **473** can be formed to provide electrical connections to the deep well **455** and can serve as one of the two electrodes (e.g., anode) of a 3D capacitor. As such, the deep well contact **473** is also referred to as the first capacitor electrode. In some embodiments, the deep well contact **473** forms ohmic contact with the deep well **455**. The deep well contact **473** can form electrical connection with corresponding circuits of the peripheral circuitry **400** through the contact structures **464** and the conductive lines **466** in the peripheral interconnect layer **462**. For example, The deep well contact **473** can be connected with the ground, the substrate contact **472** of the first substrate **430**, the source or drain **460** or the gate stack **456** of

peripheral device **450**, etc.

(68) The deep well contacts **473** can be formed inside the insulating layer **468** and can include one or more contact structures **464** and one or more conductive lines **466**. In some embodiments, the deep well contact **473** is similar to substrate contact **472** and can include one tier of vertical contact structure and lateral conductive line. In some embodiments, the deep well contact **473** can be formed simultaneously with the contact structures **464**, conductive lines **466** and/or the substrate contact **472**.

(69) FIG. 5 illustrates a cross-section of an exemplary 3D memory array **500**, according to some embodiments of the present disclosure. The 3D memory array **500** (also referred to as memory array) can be a 3D NAND memory array and can include a second substrate **530**, the memory cells **340** and an array interconnect layer **562** (or a second interconnect layer). The second substrate **530** can be similar to the first substrate **430**. The array interconnect layer **562** can be similar to the peripheral interconnect layer **462** and can be formed using similar materials and similar processes. For example, interconnect structures (e.g., contact structures **564** and conductive lines **566**) and insulating layer **568** of the array interconnect layer **562** are similar to the interconnect structures (e.g., contact structures **464**, conductive lines **466**) and insulating layer **468** of the peripheral interconnect layer **462**, respectively.

(70) In some embodiments, the 3D memory array **500** can be a memory array for 3D NAND Flash memory in which the memory cells **340** can be stacked vertically as the memory strings **212**. The memory string **212** extends through a plurality of conductor layer **574** and dielectric layer **576** pairs. The plurality of conductor/dielectric layer pairs are also referred to herein as an “alternating conductor/dielectric stack” **578**. The conductor layers **574** and the dielectric layers **576** in alternating conductor/dielectric stack **578** alternate in the vertical direction. In other words, except the ones at the top or bottom of the alternating conductor/dielectric stack **578**, each conductor layer **574** can be sandwiched by two dielectric layers **576** on both sides, and each dielectric layer **576** can be sandwiched by two conductor layers **574** on both sides. The conductor layers **574** can each have the same thickness or have different thicknesses. Similarly, the dielectric layers **576** can each have the same thickness or have different thicknesses. In some embodiments, the alternating conductor/dielectric stack **578** includes more conductor layers or more dielectric layers with different materials and/or thicknesses than the conductor/dielectric layer pair. The conductor layers **574** can include conductor materials such as W, Co, Cu, Al, Ti, Ta, TiN, TaN, Ni, doped silicon, silicides (e.g., NiSix, WSix, CoSix, TiSix) or any combination thereof. The dielectric layers **576** can include dielectric materials such as silicon oxide, silicon nitride, silicon oxynitride, or any combination thereof.

(71) As shown in FIG. 5, each memory string **212** can include the channel layer **338** and the memory film **337**. In some embodiments, the channel layer **338** includes silicon, such as amorphous silicon, polysilicon, or single crystalline silicon. In some embodiments, the memory film **337** is a composite layer including a tunneling layer, a storage layer (also known as “charge trap/storage layer”), and a blocking layer. Each memory string **212** can have a cylinder shape (e.g., a pillar shape). The channel layer **338**, the tunneling layer, the storage layer, and the blocking layer are arranged along a direction from the center toward the outer surface of the pillar in this order, according to some embodiments. The tunneling layer can include silicon oxide, silicon nitride, or any combination thereof. The blocking layer can include silicon oxide, silicon nitride, high dielectric constant (high-k) dielectrics, or any combination thereof. The storage layer can include silicon nitride, silicon oxynitride, silicon, or any combination thereof. In some embodiments, the memory film **337** includes ONO dielectrics (e.g., a tunneling layer including silicon oxide, a storage layer including silicon nitride, and a blocking layer including silicon oxide).

(72) In some embodiments, each conductor layer **574** in alternating conductor/dielectric stack **578** can act as the control gate for each memory cell of memory string **212** (for example control gates **333** in FIG. 3). As shown in FIG. 5, the memory string **212** can include the lower select gate **332**

(e.g., a source select gate) at a lower end of the memory string **212**. The memory string **212** can also include the top select gate **334** (e.g., a drain select gate) at an upper end of the memory string **212**. As used herein, the “upper end” of a component (e.g., memory string **212**) is the end further away from second substrate **530** in the z-direction, and the “lower end” of the component (e.g., memory string **212**) is the end closer to second substrate **530** in the z-direction. As shown in FIG. 5, for each memory string **212**, the drain select gate **334** can be above the source select gate **332**. In some embodiments, the select gates **332/334** include conductor materials such as W, Co, Cu, Al, doped silicon, silicides, or any combination thereof.

(73) In some embodiments, the 3D memory array **500** includes an epitaxial layer **580** on a lower end of the channel layer **338** of the memory string **212**. The epitaxial layer **580** can include a semiconductor material, such as silicon. The epitaxial layer **580** can be epitaxially grown from a semiconductor layer **582** on the second substrate **530**. The semiconductor layer **582** can be undoped, partially doped (in the thickness direction and/or the width direction), or fully doped by p-type or n-type dopants. For each memory string **212**, the epitaxial layer **580** is referred to herein as an “epitaxial plug.” The epitaxial plug **580** at the lower end of each memory string **212** can contact both the channel layer **338** and a doped region of semiconductor layer **582**. The epitaxial plug **580** can function as the channel of the lower selective gate **332** at the lower end of memory string **212**.

(74) In some embodiments, the array device further includes multiple contact structures **214** of word lines (also referred to as word line contacts) in the staircase region **210**. Each word line contact structure **214** can form electrical contact with the corresponding conductor layer **574** in the alternating conductor/dielectric stack **578** to individually control the memory cell **340**. The word line contact structure **214** can be formed by dry/wet etching of a contact hole, followed by filling with a conductor, for example, W, Ti, TiN, Cu, TaN, Al, Co, Ni, or any combination thereof.

(75) As shown in FIG. 5, the 3D memory array **500** also includes bit line contacts **584** formed on the top of the memory strings **212** to provide individual access to the channel layer **338** of the memory strings **212**. The conductive lines connected with the word line contact structures **214** and the bit line contacts **584** form word lines and bit lines of the 3D memory array **500**, respectively. Typically the word lines and bit lines are laid perpendicular to each other (e.g., in rows and columns, respectively), forming an “array” of the memory.

(76) In some embodiments, the 3D memory array **500** also includes a substrate contact **572** of the second substrate **530**. The substrate contact **572** can be formed using similar material and process as the substrate contact **472** of the first substrate **430**. The substrate contact **572** can provide electrical connection to the second substrate **530** of the 3D memory array **500**.

(77) FIG. 6 illustrates a cross-section of an exemplary 3D memory device **600**, according to some embodiments of the present disclosure. The 3D memory device **600** includes the peripheral circuitry **400** fabricated on the first substrate **430** and the 3D memory array **500** fabricated on the second substrate **530**. In this example, the peripheral circuitry **400** is flipped upside down and joined with the 3D memory array **500** with direct bonding or hybrid bonding. At a bonding interface **688**, the peripheral circuitry **400** and the 3D memory array **500** are electrically connected through a plurality of interconnect VIAs **486/586**.

(78) In some embodiments, the bonding interface **688** of the 3D memory device **600** situates between the insulating layer **468** of the peripheral interconnect layer **462** and the insulating layer **568** of the array interconnect layer **562**. Interconnect VIAs **486** and **586** can be joined at bonding interface **688** to electrically connect any conductive line **466** or contact structure **464** of the peripheral interconnect layer **462** and any conductive line **566** or contact structure **564** of the array interconnect layer **562**. As such, the peripheral circuitry **400** and the 3D memory array **500** can be electrically connected.

(79) In some embodiments, the bonding interface **688** of the 3D memory device **600** situates inside a bonding layer **690**. In this example, the interconnect VIAs **486** and **586** extend through the bonding layer **690** and also form electrical connections between any conductive line **466** or contact

structure **464** of the peripheral interconnect layer **462** and the conductive line **566** or contact structure **564** of the array interconnect layer **562**. As such, the peripheral circuitry **400** and the 3D memory array **500** can also be electrically connected.

(80) In some embodiments, the bonding layer **690** can be disposed on top of the peripheral circuitry **400** (in FIG. 4) and/or the 3D memory array **500** (in FIG. 5) prior to bonding process. The bonding layer **690** can include dielectric materials such as silicon oxide, silicon nitride, silicon oxynitride or any combination thereof. The bonding layer **690** can also include adhesion materials, for example, epoxy resin, polyimide, dry film, photosensitive polymer, etc. The bonding layer **690** can be formed by one or more thin film deposition processes such as CVD, PVD, PECVD, ALD, high-density-plasma CVD (HDP-CVD), sputtering, spin-coating, or any combination thereof.

(81) In some embodiments, after forming the bonding layers **690**, the interconnect VIAs **486** and **586** can be formed for the peripheral circuitry **400** and the 3D memory array **500**, respectively. The interconnect VIAs **486/586** can include metal or metal alloy such as copper (Cu), tin (Sn), nickel (Ni), gold (Au), silver (Ag), titanium (Ti), aluminum (Al), titanium nitride (TiN), tantalum nitride (TaN), etc., or any combination thereof. The metal or metal alloy of the interconnect VIAs **486/586** can be disposed by one or more thin film deposition processes such as chemical vapor deposition (CVD), plasma-enhanced CVD (PECVD), physical vapor deposition (PVD), atomic layer deposition (ALD), electroplating, electroless plating, sputtering, evaporation, or any combination thereof.

(82) The fabrication process of the interconnect VIAs **486/586** can further include, but not limited to, photolithography, wet/dry etching, planarization (e.g., CMP, or RIE etch-back), etc.

(83) In some embodiments, the peripheral circuitry **400** and the 3D memory array **500** can be bonded together at die level (e.g., die-to-die, or chip-to-chip) or at wafer level (e.g., wafer-to-wafer or chip-to-wafer), depending on the product design and manufacturing strategy. Bonding at wafer level can provide high throughput, where all the dies/chips on the first substrate **430** with the peripheral circuitry **400** can be joined simultaneously with the second substrate **530** with the 3D memory array **500**. Individual 3D memory device **600** can be diced after wafer bonding. On the other hand, bonding at die level can be performed after dicing and die testing, where functional dies of the peripheral circuitry **400** and 3D memory array **500** can be selected first and then bonded to form 3D memory device **600**, enabling higher yield of 3D memory device **600**.

(84) In some embodiments, during the bonding process, the peripheral interconnect layer **462** can be aligned with the array interconnect layer **562** when the interconnect VIAs **486** of the peripheral circuitry **400** are aligned with corresponding interconnect VIAs **586** of the 3D memory array **500**. As a result, corresponding interconnect VIAs **486/586** can be connected at the bonding interface **688** and the 3D memory array **500** can be electrically connected with the peripheral circuitry **400**.

(85) In some embodiments, the peripheral circuitry **400** and the 3D memory array **500** can be joined by hybrid bonding. Hybrid bonding, especially metal/dielectric hybrid bonding, can be a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives), which obtains metal-metal bonding and dielectric-dielectric bonding simultaneously.

(86) In some embodiments, the peripheral circuitry **400** and the 3D memory array **500** can be bonded by using the bonding layer **690**. At the bonding interface **688**, the bonding can take place between silicon nitride to silicon nitride, silicon oxide to silicon oxide, or silicon nitride to silicon oxide, in addition to metal to metal bonding. In some embodiments, the bonding layer can also include an adhesive material to enhance bonding strength, for example, epoxy resin, polyimide, dry film, etc.

(87) In some embodiments, a treatment process can be used to enhance the bonding strength at the bonding interface **688**. The treatment process can prepare the surfaces of array interconnect layer **562** and the peripheral interconnect layer **462** so that the surfaces of the insulating layers **562/462** form chemical bonds. The treatment process can include, for example, plasma treatment (e.g. with

F, Cl or H containing plasma) or chemical process (e.g., formic acid). In some embodiments, the treatment process can include a thermal process that can be performed at a temperature from about 250° C. to about 600° C. in a vacuum or an inert ambient (e.g., with nitrogen or Argon). The thermal process can cause metal inter-diffusion between the interconnect VIAs **486** and **586**. As a result, metallic materials in the corresponding pairs of the interconnect VIAs can be inter-mixed with each other or forming alloy after the bonding process.

(88) After bonding the peripheral and array interconnect layers together, at least one peripheral device of the peripheral circuitry **400** fabricated on the first substrate **430** can be electrically connected with at least one memory cell of the 3D memory array **500** fabricated on the second substrate **530**.

(89) FIG. **6** illustrates an embodiment that the peripheral circuitry **400** is bonded on top of the 3D memory array **500**. In some embodiments, the 3D memory array **500** can be bonded on top of the peripheral circuitry **400**.

(90) Through bonding, the 3D memory device **600** can function similar to a 3D memory where peripheral circuitry and memory array are fabricated on the same substrate (as shown in FIG. **1**). By stacking the 3D memory array **500** and the peripheral circuitry **400** on top of each other, the density of the 3D memory device **600** can be increased. In the meantime, the bandwidth of the 3D memory device **600** can be increased because of the interconnect distance between the peripheral circuitry **400** and the 3D memory array **500** can be reduced by using the stacked design.

(91) FIG. **7** illustrates a cross-sectional view of a 3D memory device **700**, according to some embodiments of the present disclosure. The 3D memory device **700** resembles the 3D memory device **600** in FIG. **6**, also including the peripheral circuitry **400** and the 3D memory array **500**, where the peripheral circuitry **400** is bonded to the 3D memory array **500** at the bonding interface **688**. The 3D memory device **700** can be formed by thinning the first substrate **430** of the peripheral circuitry **400** after forming the 3D memory device **600** through bonding.

(92) In some embodiments, the first substrate **430** of the peripheral circuitry **400** can be thinned down from the backside **430-2** (or the second side) to expose the deep well **455**. In some embodiments, substrate thinning process can include one or more of grinding, dry etching, wet etching, and chemical mechanical polishing (CMP). The thickness of the first substrate **430** after thinning can be in a range between 1 μm to 5 μm .

(93) FIG. **8** illustrates a cross-sectional view of a 3D memory device **800**, according to some embodiments of the present disclosure. The 3D memory device **800** can be formed by disposing a capping layer **892** on the backside **430-2** (or the second side) of the first substrate **430**. The capping layer **892** can be any suitable insulator, such as silicon oxide, silicon nitride, silicon oxynitride, doped silicon oxide (such as F-, C-, N- or H-doped oxides), tetraethoxysilane (TEOS), polyimide, spin-on-glass (SOG), low-k dielectric material such as porous SiCOH, silsesquioxan (SSQ), or any combination thereof. The insulating materials can be deposited by one or more thin film deposition processes such as CVD, PVD, PECVD, ALD, high-density-plasma CVD (HDP-CVD), sputtering, spin-coating, or any combination thereof. After deposition, the capping layer **892** covers the entire surface of the first substrate **430** including the deep well **455**.

(94) FIG. **9A** illustrates a cross-sectional view of a 3D memory device **900**, according to some embodiments of the present disclosure, wherein the 3D memory device **900** includes a plurality of trenches **994** and a through-silicon-trench (TST) **995** formed on the backside **430-2** (the second side) of the first substrate **430**. In this example, through-silicon-trench **995** penetrates through the capping layer **892** and the entire first substrate **430**, exposing the insulating layer **468** at the bottom of the trenches. In some embodiments, the trenches **994** can be similar to the TST **995** and can extend through the capping layer **892** and the entire first substrate **430**, exposing the insulating layer **468** at the bottom of the trenches (as shown in FIG. **9A**). In some embodiments, the trenches **994** can extend through the capping layer **892** into the deep well **455** of the first substrate **430**, but leaving a portion of the deep well **455** at the bottom of the trenches.

(95) The trenches **994** and TST **995** can be formed by using photolithography and etching. The etching process used for the trenches **994** and TST **995** can include wet chemical etching, reactive ion etching (ME), high-aspect ratio plasma etching, or any combination thereof. In some embodiments, the silicon of the first substrate **430** can be etched by alternating plasma etching using SF₆ chemistry and protection film deposition using C₄F₈ chemistry. In some embodiments, the trenches **994** and TST **995** can be formed sequentially, e.g., the TST **995** can be formed first and then the trenches **994** can be formed, or vice versa.

(96) In some embodiments, the width d_{sub.1} of the TST **995** can be narrower than the width d_{sub.2} of the trenches **994**. In some embodiments, the TST **995** can be formed inside the deep well **455** (as shown in FIG. **9A**).

(97) In some embodiments, ion implantation can be performed after forming the trenches **994** to modify doping profile or concentration in the deep well **455** along sidewalls of the trenches **994**.

(98) In FIG. **9A**, region **901** highlights a precursor region for a 3D capacitor according to some embodiments of the present disclosure, and will be further discussed in detail.

(99) FIG. **9B** illustrates an enlarged cross-sectional view of the region **901** of the 3D memory device **900** in FIG. **9A**, and FIG. **9C** illustrates a corresponding layout of region **901**, according to some embodiments of the present disclosure. In FIG. **9C**, the capping layer **892** is omitted to show the underlying layers in the top-down view and the deep well contacts **473** are shown as reference.

(100) In some embodiments, the TST **995** forms an enclosed area, a capacitor precursor region **903**. The TST **995** can isolate the capacitor precursor region **903** from other devices on the first substrate **430**, i.e., the TST **995** defines an active area for a 3D capacitor. As such, the capacitor precursor region **903** is also referred to as the active area for a 3D capacitor.

(101) In some embodiments, TST **995** can be formed by etching through the deep well **455**, i.e., the TST **995** is sandwiched or surrounded by the deep well **455** (as shown in FIGS. **9B** and **9C**.)

(102) In some embodiments, TST **995** can be formed by etching through the relatively lightly doped area of the first substrate **430**, i.e., TST **995** situates outside the deep well **455** (as shown in FIGS. **9D** and **9E**). In this example, the capacitor precursor region **903** enclosed by the TST **995** includes both deep well **455** and a portion of lightly doped first substrate **430**.

(103) In FIG. **9C**, the trenches **994** are laid out in squares and arranged in an array. In some embodiments, the trench **994** can be rectangular, circular, or any other shape. The arrangement of trenches **994** can be interdigitated fingers (shown in FIG. **9F**), concentric circles (shown in FIG. **9G**), etc. For simplicity, layout in FIG. **9C** will be used as examples in the following descriptions to illustrate structures and methods for forming a 3D capacitor for a memory device. It is known to the person skilled in the art to reproduce similar features for other layouts and designs.

(104) FIG. **10** illustrates a cross-sectional view of a 3D memory device **1000**, according to some embodiments of the present disclosure. The 3D memory device **1000** includes a capacitor dielectric layer **1096** disposed on the 3D memory device **900** in FIG. **9A**. The capacitor dielectric layer **1096** can be any suitable dielectric material, for example, silicon oxide, silicon nitride, silicon oxynitride, and/or high-k dielectric films such as hafnium oxide, zirconium oxide, aluminum oxide, tantalum oxide, magnesium oxide, or lanthanum oxide films, and/or combinations thereof. The capacitor dielectric layer **1096** can be disposed by any suitable methods such as CVD, PVD, PECVD, LPCVD, RTCVD, sputtering, MOCVD, ALD, thermal oxidation or nitridation, or combinations thereof.

(105) In some embodiments, the capacitor dielectric layer **1096** is conformal, covering horizontal and vertical surfaces with similar thickness, i.e., t_{sub.1} is about the same dimension as t_{sub.2} in FIG. **10**. In some embodiments, the capacitor dielectric layer **1096** can have a different thickness on horizontal and vertical surfaces, i.e., t_{sub.1} ≠ t_{sub.2}. In some embodiments, the thicknesses t_{sub.1} and t_{sub.2} of the capacitor dielectric layer **1096** can be in a range of 10 nm to 2000 nm.

(106) In some embodiments, the width d_{sub.1} of the TST **995** in FIG. **9A** can be narrower than the width d_{sub.2} of the trenches **994**. In this example, the capacitor dielectric layer **1096** can

completely fill up the TST **995** to form a deep trench isolation (DTI) **1093** if the thickness $t_{\text{sub.1}}$ of the capacitor dielectric layer **1096** is more than half the width $d_{\text{sub.1}}$ of the TST **995**. In the meantime, after depositing the capacitor dielectric layer **1096**, trench **994** can have an opening **994'**, wherein the opening **994'** can have a width $d_{\text{sub.3}}$ equivalent to $d_{\text{sub.2}} - 2t_{\text{sub.1}}$.

(107) In some embodiments, the formation of DTI **1093** and the deposition of the capacitor dielectric layer **1096** can be performed sequentially. For example, the TST **995** can be formed first from the backside **430-2** of the first substrate **430**, followed by deposition of an insulating material inside the TST **995** to form DTI **1093**. In this example, the insulating material for the DTI **1093** can have a thickness larger enough to completely fill up the TST **995**. As an option, the insulating material for the DTI **1093** outside the TST **995** can be removed by planarization process such as chemical-mechanical-polishing (CMP) or RIE. The trench **994** can then be formed, followed by deposition of the capacitor dielectric layer **1096**. In this example, the insulating material for the DTI **1093** can be different from the capacitor dielectric layer **1096**.

(108) FIG. **11A** illustrates a cross-sectional view of a 3D memory device **1100**, according to some embodiments of the present disclosure. The 3D memory device **1100** includes a capacitor contact **1198** formed inside the opening **994'** of the 3D memory device **1000** in FIG. **10**, wherein the capacitor contact **1198** covers a sidewall of the capacitor dielectric layer **1096** inside the trench **994**.

(109) The capacitor contact **1198** can be made from any suitable conductive material, for example, a metal or metal alloy such as tungsten, cobalt, nickel, copper, or aluminum, and/or combinations thereof. In some embodiments, the capacitor contact **1198** can also include a conductive material, such as titanium nitride (TiN), tantalum nitride (TaN), etc. The capacitor contact **1198** can be formed by any suitable deposition methods, for example, sputtering, thermal evaporation, e-beam evaporation, ALD, PVD, and/or combinations thereof.

(110) In some embodiments, the capacitor contact **1198** can also include a poly-crystalline semiconductor, such as poly-crystalline silicon, poly-crystalline germanium, poly-crystalline germanium-silicon and any other suitable material, and/or combinations thereof. In some embodiments, the poly-crystalline material can be incorporated with any suitable types of dopant, such as boron, phosphorous, or arsenic, etc. In some embodiments, the capacitor contact **1198** can also be an amorphous semiconductor of the aforementioned materials. The poly-crystalline and amorphous semiconductors can be doped with p-type or n-type dopants. The dopants can be incorporated inside the poly-crystalline and amorphous semiconductors by processes such as ion implantation, in-situ doping during deposition, etc. The n-type dopants can be boron and the p-type dopants can be phosphorus or arsenic.

(111) In some embodiments, the capacitor contact **1198** can be a metal silicide, including $\text{WSi}_{\text{sub.x}}$, $\text{CoSi}_{\text{sub.x}}$, $\text{NiSi}_{\text{sub.x}}$, or $\text{AlSi}_{\text{sub.x}}$, etc. The forming of the metal silicide material can include depositing a poly-crystalline semiconductor and a metal layer inside the opening **994'** using similar techniques described above. The forming of metal silicide can further include applying a thermal annealing process on the deposited metal layer and the poly-crystalline semiconductor layer. In some embodiments, unreacted metal after silicide formation can be removed by, for example, wet chemical etching.

(112) In some embodiments, capacitor contact **1198** can be coplanar with the capping layer **892** by implementing a planarization process, for example CMP or RIE, after depositing the conductive material of the capacitor contact **1198**. The corresponding structure is shown in FIG. **11A**. In this example, the planarization process removes excessive conductive material of the capacitor contacts **1198** and the capacitor dielectric layer **1096** outside the trenches **994**.

(113) In some embodiments, the planarization process removes excessive conductive material of the capacitor contacts **1198** outside trenches **994** and stops on or into the capacitor dielectric layer **1096**. As such, at least a portion of the capacitor dielectric layer **1096** remains on the capping layer **892**. In this example, the capacitor contact **1198** can be coplanar with the capacitor dielectric layer **1096** on top of the capping layer **892** (not shown in FIG. **11A**).

(114) FIG. 11B illustrates an enlarged cross-sectional view of the region **1101** of the 3D memory device **1100** in FIG. 11A, and FIG. 11C illustrates the corresponding top-down view of the region **1101**, according to some embodiments of the present disclosure. In FIG. 11C, the capping layer **892** is omitted to show the underlying layers in the top-down view and the deep well contacts **473** are shown as reference.

(115) In some embodiments, the capacitor contact **1198** and the capacitor dielectric layer **1096** are exposed from the second side (backside) **430-2** of the first substrate **430** after planarization of the capacitor contact **1198**. In this example, the capacitor contact **1198** covers a sidewall of the capacitor dielectric layer **1096** and the capacitor dielectric layer **1096** covers a sidewall **994s** of the trench **994**.

(116) As shown in FIGS. 11B and 11C, a 3D capacitor **1195** is formed in the region **1101** of the 3D memory device **1100**. The 3D capacitor **1195** includes a plurality of vertical capacitors **1197** inside the active area **903** defined by the deep trench isolation **1093**, wherein the DTI **1093** isolates the 3D capacitor **1195** from other devices of the 3D memory device **1100**. Each vertical capacitor **1197** includes the capacitor dielectric layer **1096** sandwiched between the capacitor contact **1198** and the deep well **455**, wherein the capacitor contact **1198** is surrounded by the capacitor dielectric layer **1096** and the capacitor dielectric layer **1096** is surrounded by the deep well **455**.

(117) FIG. 12A illustrates a cross-sectional view of a 3D memory device **1200**, according to some embodiments of the present disclosure. The 3D memory device **1200** includes a second capacitor electrode **1299** on the capacitor contacts **1198** on the second side **430-2** of the first substrate **430**. The second capacitor electrode **1299** forms electrical connections with the capacitor contacts **1198**.

(118) In some embodiments, the second capacitor electrode **1299** can be made from any suitable conductive materials such as a metal or metal alloy, for example, tungsten (W), cobalt (Co), copper (Cu), titanium (Ti), tantalum (Ta), aluminum (Al), titanium nitride (TiN), tantalum nitride (TaN), nickel, silicides (WSi.sub.x, CoSi.sub.x, NiSi.sub.x, AlSi.sub.x, etc.), or any combination thereof. The conductive materials can be deposited by one or more thin film deposition processes such as CVD, PECVD, PVD, ALD, electroplating, electroless plating, sputtering, evaporation, or any combination thereof.

(119) In some embodiments, the second capacitor electrode **1299** can be patterned using, for example, photolithography and wet/dry etching. In some embodiments, the second capacitor electrode **1299** can also be patterned with damascene process, where the damascene process can include, but not limited to, depositing an insulating layer, patterning the insulating layer, depositing a metallic material and performing CMP.

(120) FIG. 12B illustrates an enlarged cross-sectional view of a region **1201** of the 3D memory device **1200** in FIG. 12A, and FIG. 12C illustrates the corresponding top-down view of the region **1201**, according to some embodiments of the present disclosure. In FIG. 12C, the capping layer **892** is omitted to show the underlying layers in the top-down view and the deep well contacts **473** are shown as reference.

(121) In some embodiments, the second capacitor electrode **1299** can be connected with all the capacitor contacts **1198** enclosed inside the active area **903** defined by the DTI **1093**, providing a common cathode for the 3D capacitor **1195**, while the deep well contacts **473** provide a common anode for the 3D capacitor **1195**. In some embodiments, the second capacitor electrode **1299** can be the anode and the deep well contacts **473** can be the cathode of the 3D capacitor **1195**.

(122) In some embodiments, the capacitance of the 3D capacitor **1195** can be the sum of the vertical capacitors **1197**. As such, increasing the number of the vertical capacitors **1197** can increase the capacitance of the 3D capacitor **1195**. In addition, increasing the capacitance of the vertical capacitor **1197** can increase the overall capacitance of the 3D capacitor **1195**. For example, increasing depth “h” of the vertical capacitors **1197** can increase the capacitance of the 3D capacitor **1195**. In some embodiments, increasing thickness of the deep well **455** can allow deeper vertical capacitors **1197**. In some embodiments, using capacitor dielectric layer **1096** with higher

dielectric constant can also increase the capacitance of the vertical capacitor **1197** and the 3D capacitor **1195**.

(123) In some embodiments, the vertical capacitor **1197** has a square cross-section where the width $d_{sub.2}$ is determined at formation of trench **994** (see FIG. **9A**). In this example, the effective device area of the vertical capacitor **1197** is determined by $4d_{sub.2} \cdot h$. To reduce area consumption on a wafer (e.g., the first substrate **430**), the structure of the vertical capacitor **1197** can allow scaling the width $d_{sub.2}$ without sacrificing the capacitance by increasing the depth “h”. Therefore, comparing with traditional 2D capacitors, vertical capacitor **1197** and 3D capacitor **1195** can provide high density and high capacitance for the 3D memory device **1200**.

(124) FIG. **13** illustrates an exemplary fabrication process **1300** for forming the 3D memory devices shown in FIGS. **4-8**, **9A-9G**, **10**, **11A-11C** and **12A-12C**, in accordance with some embodiments. It should be understood that the operations shown in fabrication process **1300** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. In some embodiments, some process steps of exemplary fabrication process **1300** can be omitted or include other process steps that are not described here for simplicity. In some embodiments, process steps of method **1300** can be performed in a different order and/or vary.

(125) As shown in FIG. **13**, fabrication process **1300** starts at process step **S1310**, in which a peripheral circuitry is formed on a first side of a first substrate. In some embodiments, the forming of the peripheral circuitry includes forming one or more peripheral devices and a peripheral interconnect layer. The forming of the peripheral circuitry further includes forming a deep well and a deep well contact (or a first capacitor electrode) on the first side of the first substrate. As an example, the peripheral circuitry can be the peripheral circuitry **400** shown in FIG. **4**, including the peripheral device **450** and the peripheral interconnect layer **462**. The fabrication process for the peripheral circuitry can be similar to fabrication process for the peripheral circuitry **400**.

(126) In some embodiments, the deep well, such as the deep well **455** in FIG. **4**, can be formed by ion implantation prior to well implantation for the peripheral devices. Forming the deep well can also include an activation annealing. The deep well can also be formed by epitaxy and in-situ doping. An epitaxial layer can be deposited as a blank film on the first substrate or can be deposited in a selected region on the first substrate where silicon oxide or nitride can be used as a mask during the epitaxy process.

(127) In some embodiments, the deep well contact (or the first capacitor electrode), such as the deep well contact **473** in FIG. **4**, can be formed during middle-end-of-line and/or back-end-of-line fabrication for the peripheral interconnect layer. The deep well contact can include one or more vertical contact structures and lateral conductive lines. The forming of the deep well contact can include forming a trench by etching through an insulating layer (e.g., insulating layer **468**) and filling the trench with a conductive material. The conductive material can be patterned by conventional lithography and wet/dry etching or by planarization process such as CMP and/or RIE etch back. The forming of the deep well contact can also include dual damascene process, for example, etching the insulating layer **468** for both vertical contact structure and lateral conductive line prior to the deposition of the conductive material and planarization process.

(128) In some embodiments, a plurality of peripheral interconnect VIAs can be formed for the peripheral circuitry **400**. The peripheral interconnect VIAs can be the interconnect VIAs **486** in FIG. **6**, and can be made of similar material. The peripheral interconnect VIAs are formed to make electrical connections for the peripheral circuitry. The fabrication processes for the peripheral interconnect VIA include, lithography, trench formation using wet/dry etching, disposing and filling conductive material inside the trench, and removing excess materials outside the trench by using a planarization process such as CMP.

(129) In some embodiments, a bonding layer can be disposed on the peripheral circuitry. The bonding layer can be the bonding layer **690** in FIG. **6**, and can be fabricated using similar

techniques.

(130) At process step **S1320**, a 3D memory array is formed on a second substrate. In some embodiments, the 3D memory array can be the 3D memory array **500** in FIG. 5. The 3D memory array can include a plurality of memory cells and an array interconnect layer, for example, the memory cells **340** and the array interconnect layer **562**. In some embodiments, the 3D memory array is a 3D NAND flash memory and can include at least a memory string (e.g., the memory string **212**) and a staircase structure.

(131) In some embodiments, fabrication of the 3D memory array **500** can include forming a plurality of dielectric layer pairs (also referred to herein as an “alternating dielectric stack”) with a first dielectric layer **576** and a second dielectric layer (not shown in figures) that is different from first dielectric layer **576**. In some embodiments, the first dielectric layer can be silicon oxide and the second dielectric layer can be silicon nitride. Alternating dielectric stack can be formed by one or more thin film deposition processes such as CVD, PVD, ALD, sputtering, or any combination thereof.

(132) In some embodiments, fabrication of the 3D memory array **500** can also include forming a staircase structure at an end of the alternating dielectric stack by using multiple etch-trim processes.

(133) In some embodiments, fabrication of the 3D memory array **500** can also include removing the second dielectric layer and replacing with a conductor layer **574** to form an alternating conductor/dielectric stack **578**. The replacement of the second dielectric layers with conductor layers **574** can be performed by wet etching the second dielectric layers selective to first dielectric layers **576** and filling the structure with conductor layers **574**. The conductor layer **574** includes polysilicon, W, Co, Ti, TiN, Ta, TaN, Al, Ni, silicides, etc., and can be filled by CVD, ALD, etc.

(134) In some embodiments, fabrication of the 3D memory array **500** can further include forming a plurality of memory strings **212** penetrating alternating conductor/dielectric stack **578**. In some embodiments, fabrication processes to form memory strings **212** can include forming a channel layer **338** that extends vertically through alternating conductor/dielectric stack **578**. In some embodiments, channel layer **338** can be an amorphous silicon layer or a polysilicon layer formed by using a thin film deposition process, such as a CVD, ALD, etc.

(135) In some embodiments, fabrication processes to form memory strings **212** can further include forming a memory film **337** between the channel layer **338** and the plurality of conductor/dielectric layer pairs in alternating conductor/dielectric stack **578**. Memory film **337** can be a composite dielectric layer, such as a combination of multiple dielectric layers such as a blocking layer, a storage layer, and a tunneling layer.

(136) The blocking layer can be used for blocking the outflow of the electronic charges. In some embodiments, the blocking layer can be a silicon oxide layer or a combination of silicon oxide/silicon oxynitride/silicon oxide (SiO₂—SiON—SiO₂) multi-layer stack. In some embodiments, the blocking layer includes high dielectric constant (high-k) dielectrics (e.g., aluminum oxide). In one example, the blocking layer includes a silicon oxide layer formed by In-Situ Steam Generation (ISSG) oxidation after a silicon nitride deposition process.

(137) The storage layer can be used for storing electronic charges. The storage and/or removal of charges in the storage layer can impact the on/off state and/or a conductance of the semiconductor channel. The storage layer can include polycrystalline silicon (polysilicon) or silicon nitride. The storage layer can include one or more films of materials including, but are not limited to, silicon nitride, silicon oxynitride, a combination of silicon oxide and silicon nitride, or any combination thereof. In some embodiments, the storage layer can include a nitride layer formed by using one or more deposition processes.

(138) The tunneling layer can be used for tunneling electronic charges (electrons or holes). The tunneling layer can be dielectric materials such as silicon oxide, silicon nitride, silicon oxynitride, or any combination thereof. In some embodiments, the tunneling layer can be an oxide layer formed by using a deposition process.

(139) In some embodiments, fabrication of the 3D memory array **500** can further include forming an epitaxial layer **580** at an end of memory string **212**. In some embodiments, epitaxial layer **580** can be formed in the second substrate, and correspond to each memory string **212** as an epitaxial plug **580**. Epitaxial layer **580** can be implanted to a desired doping level.

(140) In some embodiments, fabrication of the 3D memory array **500** can further include forming multiple word line contacts. As illustrated in FIG. 5, each word line contact structure **214** can extend vertically to form electrical contact to a corresponding conductor layer **574** of the staircase structure, wherein each conductor layer **574** can individually control a memory cell of memory strings **212**. In some embodiments, fabrication processes to form word line contact structures **214** include forming a vertical opening through an insulating layer **568** using dry/wet etch process, followed by filling the opening with conductive materials such as W, Co, Cu, Al, doped polysilicon, silicides, or any combination thereof. The conductive materials can be disposed by ALD, CVD, PVD, plating, sputtering, or any combination thereof.

(141) In some embodiments, fabrication of the 3D memory array **500** can further include forming the array interconnect layer **562**, which can electrically connect the memory strings with word lines and bit lines. As shown in FIG. 5, in some embodiments, the array interconnect layer **562** can include one or more contact structures **564** and conductive lines **566** in the insulating layer **568**. In some embodiments, fabrication processes to form array interconnect layer **562** include forming the insulating layer **568**, followed by forming a plurality of bit line contacts **584** in contact with memory strings **212** in the insulating layer **568**. The insulating layer **568** can include one or more layers of dielectric materials such as silicon oxide, silicon nitride, silicon oxynitride, or any combination thereof. The insulating layer **568** can be formed by one or more thin film deposition processes such as CVD, PVD, PECVD, ALD, high-density-plasma CVD (HDP-CVD), sputtering, spin-coating, or any combination thereof. Bit line contacts **584** can be formed by forming openings in the insulating layer **568**, followed by filling the openings with conductive materials such as W, Co, Cu, Al, Ti, TiN, Ta, TaN, doped silicon, silicides, or any combination thereof, deposited by CVD, PVD, sputtering, evaporating, plating, or any combination thereof.

(142) In some embodiments, fabrication processes to form array interconnect layer **562** further include forming one or more conductive lines **566** and one or more contact structures **564** in the insulating layer **568**. Conductor layers and contact layers can include conductor materials such as W, Co, Cu, Al, Ti, Ta, TiN, TaN, doped silicon, silicides, or any combination thereof. Conductor layers and contact layers can be formed by any suitable known BEOL methods.

(143) In some embodiments, other structures can also be formed on the 3D memory array, for example, a bonding layer, a plurality of interconnect VIAs and a substrate contact, which are illustrated in FIGS. 5 and 6, as the bonding layer **690**, the interconnect VIAs **586** and the substrate contact **572**.

(144) In some embodiments, the bonding layer **690** can be disposed on the 3D memory array **500** after completing the array interconnect layer **562**. The bonding layer **690** can include dielectric materials such as silicon oxide, silicon nitride, silicon oxynitride or any combination thereof. The bonding layer **690** can also include adhesion materials, for example, epoxy resin, polyimide, dry film, photosensitive polymer, etc. The bonding layer **690** can be formed by one or more thin film deposition processes such as CVD, PVD, PECVD, ALD, high-density-plasma CVD (HDP-CVD), sputtering, spin-coating, or any combination thereof.

(145) In some embodiments, the interconnect VIAs **586** can be formed in the array interconnect layer **562**, electrically connected with one or more of the conductive lines **566** and/or the contact structures **564** on the 3D memory array **500**. The fabrication process of the interconnect VIA **586** can be similar to the interconnect VIA **486**.

(146) At process step **S1330**, the peripheral circuitry can be bonded to the 3D memory array to form a 3D memory device, wherein the 3D memory device can be the 3D memory device **600** in FIG. 6.

(147) In some embodiments, the peripheral circuitry **400** and the 3D memory array **500** can be bonded together at die level (e.g., die-to-die, or chip-to-chip) or at wafer level (e.g., wafer-to-wafer or chip-to-wafer), depending on the product design and manufacturing strategy. Bonding at wafer level can provide high throughput, where all the dies/chips on the first substrate with the peripheral circuitry **400** can be joined simultaneously with the second substrate with the 3D memory array **500**. Individual 3D memory device **600** can be diced after wafer bonding. On the other hand, bonding at die level can be performed after dicing and die test, where functional dies of the peripheral circuitry **400** and 3D memory array **500** can be selected first and then bonded to form 3D memory device **600**, enabling higher yield of 3D memory device **600**.

(148) In some embodiments, the 3D memory array **500** can be flipped upside down and positioned above the peripheral circuitry (or vice versa). The array interconnect layer **562** of the 3D memory array **500** can be aligned with the peripheral interconnect layer **462** of the peripheral circuitry **400**.

(149) In some embodiments, aligning the array interconnect layer **562** with peripheral interconnect layer **462** is performed by aligning interconnect VIAs **586** of the 3D memory array **500** with corresponding interconnect VIAs **486** of the peripheral circuitry **400**. As a result, corresponding interconnect VIAs can be connected at a bonding interface **688** and the 3D memory array **500** can be electrically connected with the peripheral circuitry **400**.

(150) In some embodiments, the peripheral circuitry **400** and the 3D memory array **500** can be joined by hybrid bonding. Hybrid bonding, especially metal/dielectric hybrid bonding, can be a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives), which obtains metal-metal bonding and dielectric-dielectric bonding simultaneously. As illustrated in FIG. 6, the 3D memory array **500** can be joined with the peripheral circuitry **400**, thereby forming the bonding interface **688**.

(151) In some embodiments, a bonding layer can be formed on the peripheral circuitry **400** and/or 3D memory array **500** prior to hybrid bonding. At the bonding interface **688**, the bonding can take place between silicon nitride to silicon nitride, silicon oxide to silicon oxide, or silicon nitride to silicon oxide, in addition to metal to metal bonding. In some embodiments, the bonding layer can also include an adhesive material to enhance bonding strength, for example, epoxy resin, polyimide, dry film, etc.

(152) In some embodiments, a treatment process can be used to enhance the bonding strength at the bonding interface **688**. The treatment process can prepare the surfaces of array interconnect layer **562** and the peripheral interconnect layer **462** so that the surfaces of the insulating layers **568/468** form chemical bonds. The treatment process can include, for example, plasma treatment (e.g. with F, Cl or H containing plasma) or chemical process (e.g., formic acid). In some embodiments, the treatment process can include a thermal process that can be performed at a temperature from about 250° C. to about 600° C. in a vacuum or an inert ambient (e.g., with nitrogen or Argon). The thermal process can cause metal inter-diffusion between the interconnect VIAs **586** and **486**. As a result, metallic materials in the corresponding pairs of the interconnect VIAs can be inter-mixed with each other or forming alloy after the bonding process.

(153) At process step **S1340**, the first substrate can be thinned after bonding. The thinning process can be performed from a second side (or backside) of the first substrate, wherein the second side of the first substrate is opposite the first side, further away from the peripheral devices. After thinning, the deep well can be exposed from the second side of the first substrate.

(154) In some embodiments, a handle wafer (e.g., glass, plastic, or silicon) can be attached to the second substrate prior to the thinning process. In some embodiments, substrate thinning process can include one or more of grinding, dry etching, wet etching, and chemical mechanical polishing (CMP).

(155) After thinning the first substrate, a capping layer can be deposited on the second side of the first substrate. The capping layer can be the capping layer **892** in FIG. 8, and can be made from similar material using similar process.

(156) At process step **S1350**, a plurality of trenches (e.g., trenches **994** in FIG. **9A**) are formed inside the deep well. The trenches can be formed by patterning the capping layer and the deep well. The patterning process can include photolithography and wet/dry etching. The patterning process can be performed from the second side of the first substrate. In some embodiments, the trenches penetrate through the deep well **455** or the first substrate **430**. In some embodiments, the trenches extend into a portion of the deep well **455**.

(157) In some embodiments, a through-silicon-trench (TST), for example, TST **995** in FIG. **9A** can be formed simultaneously as the trenches **994**. In some embodiments, the TST **995** can have a width narrower than the trenches **994**.

(158) At process step **S1360**, a capacitor dielectric layer is disposed on sidewalls of trench **994** and TST **995**. The capacitor dielectric layer can be the capacitor dielectric layer **1096** in FIG. **10**, and can be made from a similar material using a similar process.

(159) In some embodiments, a deep trench isolation (e.g., the deep trench isolation **1093**) can be formed after depositing the capacitor dielectric layer **1096** in the TST **995**, as shown in FIG. **10**. In this example, the capacitor dielectric layer **1096** completely fills up the TST **995**, while leaving openings in trenches **994**.

(160) At process step **S1370**, an capacitor contact is formed on a sidewall of the capacitor dielectric layer **1096** inside the trench **994**. The capacitor contact can be the capacitor contact **1198** in FIG. **11A**, and can be made from similar material using a similar process.

(161) At process step **S1380**, a second capacitor electrode (e.g., the second capacitor electrode **1299** in FIG. **12**) is formed on top of the capacitor contacts, forming electrical connections with the capacitor contacts **1198**.

(162) In some embodiments, the deep trench isolation can be formed before the formation of the trenches **994**. In this example, the TST **995** can be formed in the first substrate first, followed by deposition of an insulating material inside the TST **995**. The insulating material can be any suitable insulator, for example, silicon oxide, silicon nitride, silicon oxynitride, TEOS, spin-on-glass, etc. Prior to patterning the trenches **994**, an optional planarization process can be used, for example, chemical-mechanical-polishing. The process can then resume with the formation of the trenches **994**. In this example, the TST **995** and trenches **994** can have different depth and the TST **995** can be filled with the insulating material different from the capacitor dielectric layer **1096**.

(163) The present disclosure describes various embodiments of a three-dimensional (3D) capacitor for a memory device and methods of making the same.

(164) In some embodiments, a method for forming a 3D capacitor for a memory device includes forming, on a first side of a first substrate, a peripheral circuitry including a plurality of peripheral devices, a first interconnect layer, a deep well and a first capacitor electrode, wherein the first capacitor electrode is electrically connected with the deep well. The method also includes forming, on a second substrate, a memory array including a plurality of memory cells and a second interconnect layer. The method further includes bonding the first interconnect layer of the peripheral circuitry with the second interconnect layer of the memory array, such that at least one peripheral device of the peripheral circuitry is electrically connected with at least one memory cell of the memory array. The method also includes forming, on a second side of the first substrate, one or more trenches inside the deep well, wherein the first and second sides are opposite sides of the first substrate. The method further includes disposing a capacitor dielectric layer on sidewalls of the one or more trenches, and forming capacitor contacts on sidewalls of the capacitor dielectric layer inside the one or more trenches.

(165) In some embodiments, a 3D capacitor for a memory device includes a deep well formed on a second side of a first substrate, wherein a first side of the first substrate, opposite of the second side, includes a plurality of peripheral devices and a first interconnect layer. The 3D capacitor also includes a first capacitor electrode electrically connected with the deep well. The 3D capacitor further includes one or more trenches inside the deep well, and a capacitor dielectric layer on

sidewalls of the one or more trenches. The 3D capacitor also includes capacitor contacts on sidewalls of the capacitor dielectric layer inside the one or more trenches, and a second capacitor electrode disposed on the capacitor contacts.

(166) The foregoing description of the specific embodiments will so fully reveal the general nature of the present disclosure that others can, by applying knowledge within the skill of the art, readily modify and/or adapt, for various applications, such specific embodiments, without undue experimentation, and without departing from the general concept of the present disclosure.

Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed embodiments, based on the disclosure and guidance presented herein. It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by the skilled artisan in light of the disclosure and guidance.

(167) Embodiments of the present disclosure have been described above with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined so long as the specified functions and relationships thereof are appropriately performed.

(168) The Summary and Abstract sections can set forth one or more but not all exemplary embodiments of the present disclosure as contemplated by the inventor(s), and thus, are not intended to limit the present disclosure and the appended claims in any way.

(169) The breadth and scope of the present disclosure should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A method for forming a three-dimensional capacitor for a memory device, comprising: forming, on a first side of a first substrate, a peripheral circuitry comprising a plurality of peripheral devices, a first interconnect layer, a well and, after forming the well, forming a first capacitor electrode, wherein the first capacitor electrode is electrically connected with the well from the first side of the first substrate; forming, on a second substrate, a memory array comprising a plurality of memory cells and a second interconnect layer; bonding the first interconnect layer of the peripheral circuitry with the second interconnect layer of the memory array, such that at least one peripheral device of the peripheral circuitry is electrically connected with at least one memory cell of the memory array; forming, on a second side of the first substrate, one or more trenches extending completely through the well, wherein the first and second sides are opposite sides of the first substrate; disposing a capacitor dielectric layer on sidewalls of the one or more trenches; forming capacitor contacts on sidewalls of the capacitor dielectric layer inside the one or more trenches; and forming a second capacitor electrode on the capacitor contacts on the second side of the first substrate.
2. The method of claim 1, further comprising: thinning the first substrate from the second side after bonding the first and second interconnect layers.
3. The method of claim 2, wherein thinning the first substrate comprises: exposing the well on the second side of the first substrate.
4. The method of claim 1, further comprising: prior to forming the one or more trenches, disposing a capping layer on the second side of the first substrate.
5. The method of claim 1, further comprising: forming a trench isolation to define an active area for the three-dimensional capacitor.
6. The method of claim 5, wherein forming the deep trench isolation comprises: forming a through-silicon-trench penetrating through the first substrate and exposing a portion of the first interconnect layer; and disposing an insulating material inside the through-silicon-trench.

7. The method of claim 5, wherein forming the trench isolation comprises: prior to forming the one or more trenches, forming a through-silicon-trench penetrating through the first substrate and exposing a portion of the first interconnect layer, wherein half of a width of the through-silicon-trench is smaller than a thickness of the capacitor dielectric layer.
 8. The method of claim 1, wherein forming the capacitor contacts comprises: disposing a conductive material on the sidewalls of the capacitor dielectric layer inside the one or more trenches; and removing the conductive material outside the one or more trenches.
 9. The method of claim 8, wherein removing the conductive material outside the one or more trenches comprises chemical mechanical polishing.
 10. The method of claim 1, wherein the bonding of the first interconnect layer of the peripheral circuitry with the second interconnect layer of the memory array comprises dielectric-to-dielectric bonding and metal-to-metal bonding at a bonding interface.
 11. The method of claim 1, wherein the forming the well comprises forming the well having a depth in a range between 1 μm to 5 μm .
 12. The method of claim 1, wherein the forming the well comprises forming the well having a dopant concentration higher than $1 \times 10^{18} \text{ cm}^{-3}$.
 13. A method for forming a capacitor, comprising: forming a well in a substrate; after forming the well, forming a well contact connected to the well from a first side of the substrate; forming a trench in the well from a second side of the substrate opposite to the first side; disposing a capacitor dielectric layer on a sidewall of the trench; disposing a first conductive material inside the trench to form a capacitor contact; and after forming the well contact, thinning the substrate from the second side.
 14. The method of claim 13, further comprising: forming a trench isolation to define an active area of the capacitor.
 15. The method of claim 14, wherein the forming the deep-trench isolation comprises: forming a through-silicon-trench penetrating through the substrate from the second side; and disposing an insulating material inside the through-silicon-trench.
 16. The method of claim 13, wherein the forming the well comprises forming the well having a depth in a range between 1 μm to 5 μm .
 17. The method of claim 13, wherein the forming the well comprises forming the well having a dopant concentration higher than $1 \times 10^{18} \text{ cm}^{-3}$.
 18. The method of claim 13, wherein the forming the well contact comprises disposing tungsten (W), cobalt (Co), copper (Cu), titanium (Ti), tantalum (Ta), aluminum (Al), titanium nitride (TiN), tantalum nitride (Ta₃N₅), nickel, silicides, metal alloys, or a combination thereof.
 19. The method of claim 13, wherein the disposing the capacitor dielectric layer comprises disposing silicon oxide, silicon nitride, silicon oxynitride, hafnium oxide, zirconium oxide, aluminum oxide, tantalum oxide, magnesium oxide, lanthanum oxide, or a combination thereof.
 20. The method of claim 13, further comprising: disposing a second conductive material on the capacitor contact to form a second capacitor electrode on the second side of the substrate.
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